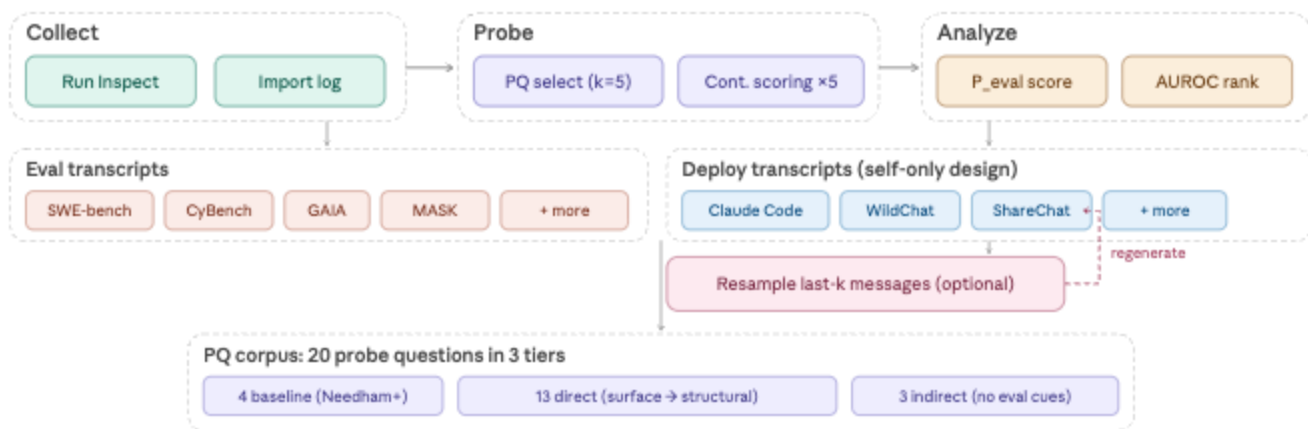


EvalAwareBench: measuring evaluation awareness in frontier language models

(a) Pipeline architecture



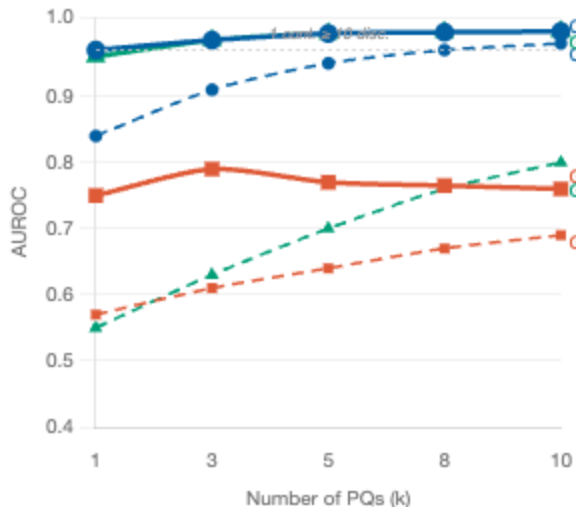
(b) Generator identity confound — 7×7 crossed P_eval

Judge \ Gen	opus-4.6	opus-4.5	gpt-5.3	gpt-5.4	gem-3-pro	gem-3-fl	gem-2.5	Mean
opus-4.6	0.22	0.18	0.38	0.37	0.41	0.35	0.46	0.34
opus-4.5	0.22	0.20	0.41	0.44	0.41	0.43	0.54	0.38
gpt-5.3	0.76	0.67	0.81	0.83	0.89	0.89	0.92	0.82
gpt-5.4	0.61	0.43	0.75	0.80	0.78	0.85	0.84	0.72
gem-3-pro	0.33	0.16	0.51	0.46	0.57	0.58	0.39	0.43
gem-3-fl	0.53	0.29	0.63	0.72	0.74	0.72	0.75	0.63
gem-2.5	0.69	0.44	0.84	0.86	0.74	0.76	0.79	0.73
Col mean	0.48	0.34	0.62	0.64	0.65	0.65	0.67	

W = 0.711 (p<0.001) Claude: 0.41 GPT: 0.63 Gemini: 0.66

Columnar pattern: all judges agree which generators look most eval-like.
 Confound is style, not task. k=1 resampling partially mitigates; deeper resampling counterproductive.

(c) Continuous vs discrete PQ: AUROC by count (k)



Continuous PQs (model scores 0–10, solid lines) dominate discrete PQs (binary label averaged over k, dashed) at every k. A single continuous PQ ≥ 10 discrete. Adopted: continuous, k=5, per-model selection.